

ARM CORTEX Processors

↳ Introduction to ARM CORTEX series:

- The ARM CORTEX series represents a significant advancement over the classical ARM processor series, offering improved performance, power efficiency, scalability & security.
- cortex processors are widely used in smartphones, tablets & embedded systems.

i) cortex A (Application) series:

Designed for high performance applications including mobile devices, tablets, & complex consumer electronics.

ii) cortex R (Real Time) series:

For real time applications requiring high reliability & deterministic behaviour, such as automotive systems & industrial control.

iii) cortex M (Microcontroller) series:

for low power & cost sensitive embedded applications like microcontrollers for IoT devices.

↳ Advantages over classical series:

ARM cortex series offers several advantages over classical arm processor series (like ARM7, ARM9, ARM11) that includes.

i) Performance Efficiency:

- Advanced microarchitectures with enhanced instruction sets that improve performance.
- Higher level of degree of instruction-level parallelism.

i) Power efficiency:

- ARM cortex cores are designed with power efficiency in mind, using features like low power idle states, clock gating & dynamic voltage.
- The cortex M series is optimized for ultra low power consumption making it ideal for battery operated devices.

ii) Scalability:

- offer wide range of options from high performance (cortex A) to low power microcontrollers (cortex M).

iii) Enhanced security:

- Provides security features like ARM TrustZone.

iv) Advanced debugging & Development Tools:

- ARM cortex processors are supported by rich ecosystem of development tools & software.

v) Extensive ecosystem & community:

- ARM cortex series benefits from a vast ecosystem of third party tools, ~~third~~ software libraries, operating systems & large developer community.

↳ ARM cortex series for embedded system design:

i) Flexibility & versatility:

- cortex M series - low power consumption
- cortex R series - focus on real time performance
- cortex A series - offers high computational performance

ii) Integration & peripherals:

- Cortex processors offer are integrated with a wide range of peripherals & interfaces.
- These peripherals include communication interfaces (UART, I2C, SPI) Timers, ADC DAC

iii) Cost efficiency:

- ARM's wide availability of ARM based microcontrollers from multiple vendors lead to competitive pricing & wide variety of options for designers.

iv) Software support:

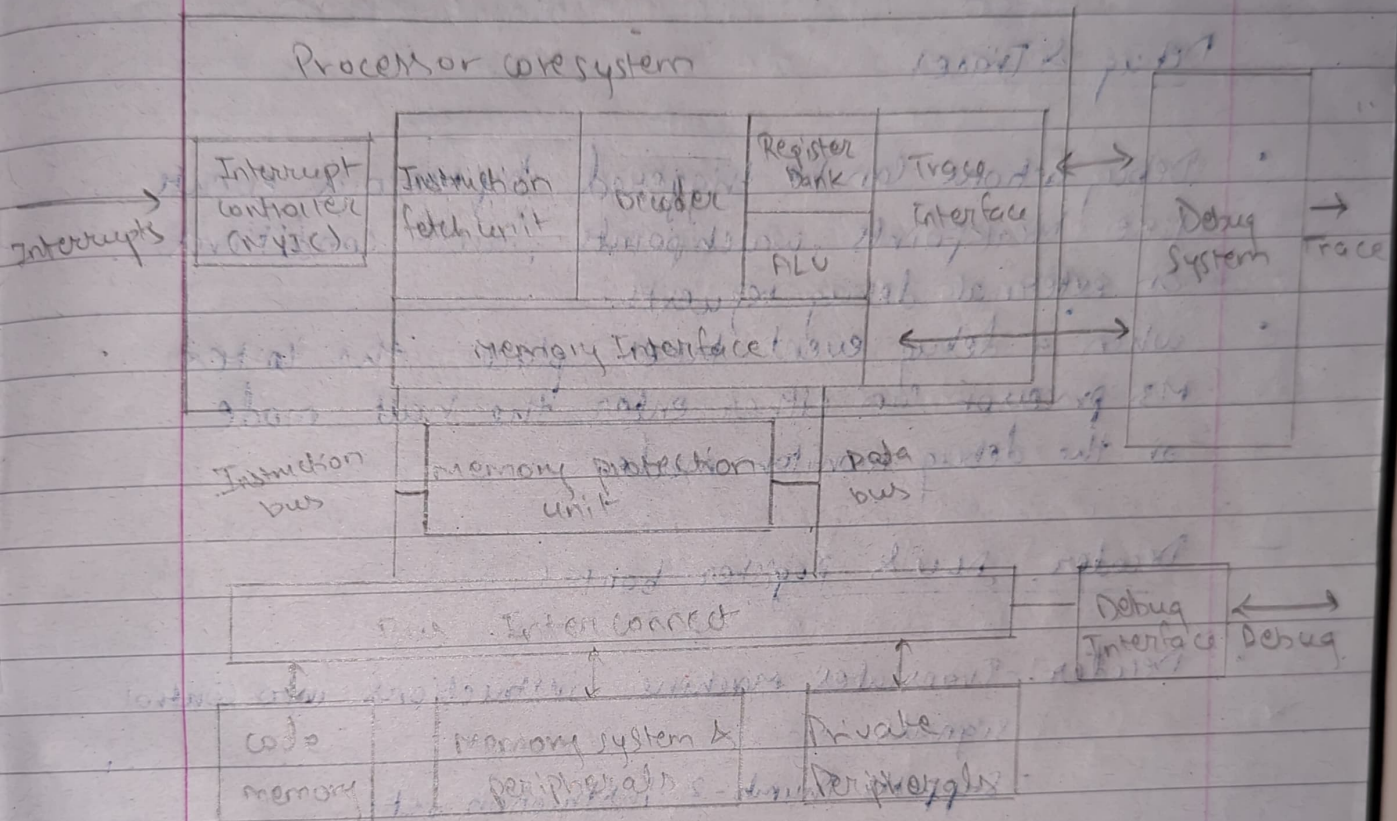
- Extensive support for real time operating systems & middleware ensures that developers can build robust & scalable software for their embedded systems.

→ Cortex A, R, M!

Aspect	Cortex A series	Cortex R series	Cortex M series
Purpose / Features	High Performance applications	Real time, safety critical applications	low power micro controllers applications
versions	Cortex-A5, A7, A9, A15, A17, etc	Cortex-R4, R5, R7, R8	Cortex-M0, M1, M2, M3, M4, M7, etc
Applications	smartphones & tablets consumer electronics Infotainment systems	Engine control safety systems Patient monitoring	IoT devices wearables Industrial control.
Power Consumption	80 $\mu\text{W}/\text{MHz}$	120 $\mu\text{W}/\text{MHz}$	8 $\mu\text{W}/\text{MHz}$
Response time	very good	Best	medium
multicore support	yes	yes	limited.
FPU	yes	yes	optional
Pipeline	long Pipeline	medium pipeline	short pipeline
clock	High	High	less relatively

→ ARM cortex M-3

• Architecture:



• Interrupt controller (NVIC)?

- The NVIC provides the processor's outstanding interrupt handling abilities.
- 32 general purpose physical interrupts.
- NVIC supports stacking of interrupts.
- Priority levels can be changed by software during run time.

• Memory protection unit

- The MPU enables the application to be broken down into a set of processes.

- The MPU separates the memory into distinct regions & implements protection by preventing disallowed accesses.

- Debug & Trace

- Debug actions can be triggered by various events like breakpoints, watchpoints, fault conditions or external debug requests.
- When a debug event takes place, the Cortex M3 processor can either enter the halt mode or the debug monitor mode.

- Decoder, ALU & Register bank

- **Decoder**:- Translates machine instructions into control signals
 - Supports Thumb-2 instruction set.
- **ALU** :- Performs arithmetic & logical operations.
 - updates condition flags based on result of operations.
- **Register Bank**:- Stores operands & results for ALU operations.
 - Special purpose registers like stack pointer, program counter & link register.

↳ Features of ARM cortex M3

- 32 bit RISC Processor.

- Thumb-2 Instruction set: combines 16 bit & 32 bit instructions for code improved code density & performance.
- Harvard Architecture: separate instruction & data buses for higher speed.
- 3 stage Pipeline - Fetch
Decode
Execute
- nested vectored Interrupt controller (NVIC): supports upto 240 interrupt lines with 256 levels of priority
- Memory Protection unit (MPU): Enhances system reliability by protecting critical regions of memory.
- low power consumption
- fast Interrupt Handling
- Robust security & Reliability.

↳ Functional Description of ARM Cortex-M3

i) Processor Core:

- 32 bit Architecture.
- Pipeline - Fetch
Decode
Execute

ii) Register Bank:

- General purpose registers:
R0-R12 : used for general data storage & operations
- Special Registers:
R13 (SP): Points to top of current stack
R14 (LR): stores return address for function calls
R15 (PC): contains address of next instruction to execute.

iii) Memory system:

- Hardware architecture
- Memory Map:
 - Code region: for storing executable region.
 - SRAM region: for data storage.
 - Peripheral region: for accessing peripheral devices.
 - External devices: optional external devices.
- Memory Protection unit (MPU):
 - for enhancing security & reliability.

iv) Bus Interfaces:

- AHB - like (Advanced High speed bus for connecting to memory & peripherals)
- APB (Advanced Peripheral Bus): used for slower peripherals & device registers.

v) Interrupt system:

- Nested vectored Interrupt Controller (NVIC):
 - Priority levels
 - nested Interrupts
 - vector Table.
 - fast response.

vi) Debug & Trace units:

- Integrated Debug module
- Instrumentation Trace macrocell
- Data watchpoint & Trace units
- Embedded Trace macrocell

vii) Power Management:

- sleep modes:
 - sleep: Reduces power while maintaining processor

State.

- Deep sleep: further reduces power with more components turned off.

• Wakeup Interrupt Controller (WIC):

Allows the processor to wake up from low power modes in response to specific interrupts.

viii) System Control Block (SCB):

- System Handlers: manages system exceptions like reset, NMI & hard faults.
- Configuration & control registers: Provides control over system behaviours.

ix) Performance Features:

- Thumb2 Instruction set
- Enhanced DSP Instructions: supports digital signal processing with efficient multiply accumulate operations

↳ Advantages of ARM Cortex M3 for embedded Applications

- High Performance & efficiency
- Low power consumption
- Deterministic Interrupt Handling
- Robust Debugging & Trace Capabilities
- Comprehensive memory Protection
- Rich ecosystem & development support
- Cost effective solution
- Security features.
- Ease of use

Comparison of ARM Cortex M3 & ARM7

Feature	ARM Cortex M3	ARM7
Architecture	ARMv7-M (Harvard)	ARMv7T (von Neuman)
Instruction set	Thumb-2 (16 bit & 32 bit)	ARM (32-bit) & Thumb (16-bit)
Performance	High	Moderate
Power consumption	low power consumption	Higher power consumption
Interrupt Handling	Nested vectored Interrupt Controller (NVIC)	vectored Interrupt Controller (VIC)
Memory Protection	MPU	No MPU
Bus Interface	AHB-Lite, APB	AHB, APB
Typical Applications	IoT, automotive, industrial control	General, embedded applications, legacy systems
Security features	Trustzone	No hardware security features
Sleep modes	None	Integrated.

↳ Firmware development Development using CMSIS standard for ARM cortex.

- The ARM ~~on~~ cortex microcontroller software Interface Standard (CMSIS) is an hardware abstraction layer for the cortex m processors.
- This standard specifies the various standards that are required for the embedded system based on cortex m processors.

- Components:

- i) CMSIS Core: Provides standardized interface for the cortex - m processors.
- ii) CMSIS Driver: It is a set of code that defines the interface drivers for the various standard peripherals like communication, display, etc.
- iii) CMSIS DSP: collection of various DSP operations.
- iv) CMSIS RTOS API: standard API for real time operating systems.
- v) CMSIS DAP: It is a debugging port for debugging the programs written onto the cortex - m based systems.
- vi) CMSIS SVD: system view description format for describing the system peripherals & their register sets.
- vii) CMSIS Pack: It describes all relevant parts of the software ~~the~~ like library, files, documentation, header files, etc.